Sushmit Arora

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LinkedIn | GitHub

Education

SRM Institute of Science and Technology — Delhi NCR, India
B.Tech in Computer Science and Engineering (AI & ML) — *Aug 2024 – 2028*
CGPA: 9.47/10 (2nd year)

Projects

Retention Decision Engine: Customer Churn Prediction — GitHub
Python, LightGBM, SHAP, FastAPI, Valkey, MLflow, Docker, Streamlit — *Public KKBox data*

- Problem: subscription businesses can't tell in advance which customers will lapse, when, or why.
- Built calibrated LightGBM and survival models that predict who will lapse and when (7- and 30-day horizons), with per-customer reason codes.
- Served scores via FastAPI + Valkey (p95 latency 69.4 ms vs. a self-set 100 ms target) with an MLflow registry, Docker Compose and a Streamlit dashboard.
- Set pass marks before seeing results and reported a 996/1,000 API-vs-offline parity miss as "not met" rather than tuning it away; ~2,900 tests.

ThermoScope: Thermal Source Classifier — Live Demo | GitHub
Python, LightGBM, NASA FIRMS, OpenStreetMap, Leaflet — *Smart India Hackathon 2026 (NTRO)*

- Problem: a satellite heat detection doesn't say what caused it (industry, flare, crop burning, wildfire).
- Grouped NASA FIRMS detections over Gujarat into 20,409 events, enriched with OpenStreetMap, power-plant and land-cover data.
- Trained a 4-class LightGBM classifier on an audited 11-feature set with a temporal split (train 2019–2022, test 2023–2025).
- Automated inference every 30 minutes (no retraining) behind a Leaflet map dashboard; 496 tests.

Legal Assistant RAG Chatbot — GitHub
Python, FastAPI, LangChain, ChromaDB, sentence-transformers, Docker

- Problem: legal answers must trace to authoritative sources, and plain LLMs can state law without one.
- Built a RAG pipeline over Indian statutes and judgments (PDF extraction with OCR fallback, ChromaDB) that answers with bracketed citations.
- Added a verifier that checks each citation maps to a retrieved passage, retries twice and flags unmatched references; FastAPI + CLI, pytest, Docker.

Experience

Technical Lead — Sep 2025 – Present
Google Developer Groups, SRMIST

- Lead the technical team; built a basic AI email automation and the frontend of the GDG web page.
- Led and hosted Prompt Rush 3.0, a hackathon organised by Google Developer Groups, SRMIST.

Technical Team Member — Jul 2025 – Mar 2026
GENESIS: Web3 Community

- Developed the landing page of the community's website.

Technical Skills

Languages: Python, SQL (basic), C++, JavaScript
Machine Learning & Data: LightGBM, scikit-learn, SHAP, Pandas, NumPy, Calibration, Survival Analysis
Generative AI & LLMs: RAG, LangChain, ChromaDB, sentence-transformers, Claude
Backend & Tools: FastAPI, React.js, Streamlit, Leaflet, Docker, MLflow, Git, GitHub Actions (CI), pytest

Achievements & Certifications

- Call2Code Hackathon (IEEE MUJ): Finalist, top 15 of 240
- TechSprint Hackathon (Google Developer Groups, SRMIST): Top 10 Finalist
- Data Science, Cisco | Data Visualisation, Tata | Solutions Architecture, AWS